Phase 1 - Corpus Summary Report

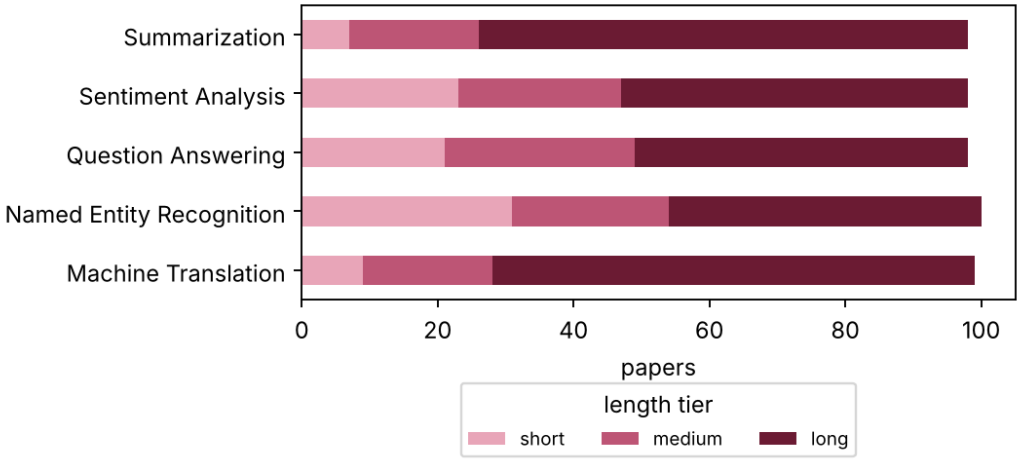
Information Retrieval - arXiv cs.CL hybrid search engine. Princess Sumaya University for Technology.

1. Overview

The corpus contains **493 papers** from arXiv category **cs.CL** (Computation and Language), labelled with **5 sub-topics**: machine translation, summarization, sentiment analysis, named entity recognition and question answering. Every document has a title, full extracted text, arXiv ID/link, sub-topic label, page count and length tier. All papers come from arXiv only.

Table 1 - papers per sub-topic and length tier

Sub-topic	Short	Medium	Long	Total
Machine Translation	9	19	71	99
Named Entity Recognition	31	23	46	100
Question Answering	21	28	49	98
Sentiment Analysis	23	24	51	98
Summarization	7	19	72	98
All	91	113	289	493



2. How the corpus was collected (reproducibility)

Script: `download_corpus.py` (arXiv Python client, then PyMuPDF to extract PDF text). For each sub-topic one query was run, sorted by **relevance**, with **max_results = 100**:

Sub-topic	arXiv API query
Machine Translation	cat:cs.CL AND "machine translation"
Summarization	cat:cs.CL AND "summarization"
Sentiment Analysis	cat:cs.CL AND "sentiment analysis"
Named Entity Recognition	cat:cs.CL AND "named entity recognition"
Question Answering	cat:cs.CL AND "question answering"

Date range: no submission-date filter was applied, so the date range is whatever the relevance ranking returned. The papers span **1999 to 2026** (derived from the arXiv IDs); the data was downloaded in early November 2026. Because relevance ranking on the live arXiv index can change over time, re-running the script later may return a slightly different set; the exact papers are therefore stored in the committed dataset

`our_arxiv_corpus.csv`.

Length tiers use the PDF page count (references and appendices included): short = 6 pages or fewer, medium = 7-9 pages, long = more than 9 pages. The assignment leaves exactly 6 pages undefined; we placed 6-page papers in *short*.

3. Cleaning applied to the raw download

The raw download had 500 rows. We removed **6** papers whose PDF text extraction failed (fewer than 3,000 characters; five had a page count of 1) and **1** duplicate (arXiv 1912.02998 matched both the machine-translation and question-answering queries; the first label was kept). We also stored a clean arXiv ID and URL without the version suffix and added the submission year. Script: `clean_corpus.py`. Final size: **493 papers**.

4. Corpus statistics

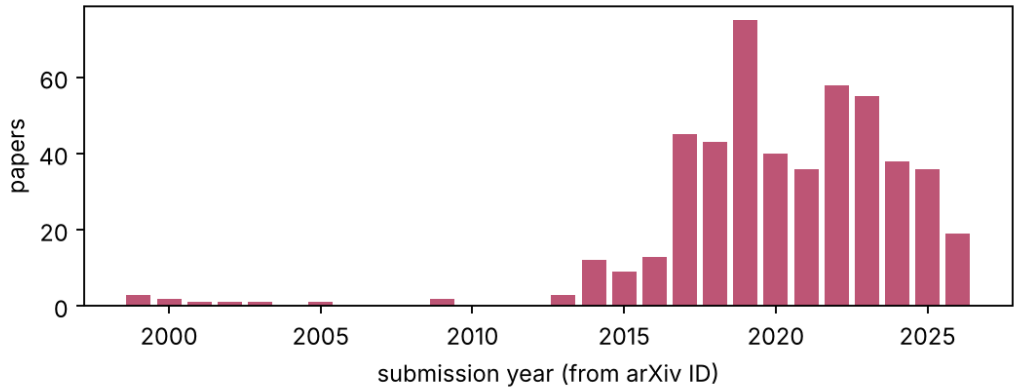
Table 2 - length statistics per tier

Tier	Papers	Pages min/median/max	Words min/median/max
short	91	2 / 5 / 6	996 / 3,247 / 5,573
medium	113	7 / 8 / 9	854 / 4,668 / 7,243
long	289	10 / 13 / 192	651 / 7,055 / 50,005

Table 3 - overall statistics

Metric	Value
Total papers	493
Total words (extracted text)	3,163,965
Mean words per paper	6,418
Median pages	10
Papers from 2020 or later	282
Papers before 2015	26

Figure 2 - publication years



5. Limitations

- **Sub-topics are balanced (about 100 per topic) but length tiers are not:** long papers are 59% of the corpus, and machine translation (9) and summarization (7) have few short papers. All three tiers are present in every topic.

- **No date filter.** The corpus mixes papers from 1999 to 2026; most are 2017 or later.

- **One outlier:** arXiv 1910.08592 has 192 pages (a thesis-length document); it is kept and affects only the maximum page count.

- **Labels come from the query** (a paper gets the topic whose query returned it) and were not hand-verified; a paper on two topics keeps one label.

- **Page count includes references and appendices**, and text extracted from PDFs contains headers, formulas and reference lists.